

Heating and Cooling: Particles and Phase Changes

PhET States of Matter · PhET

Years 7-8

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: <https://phet.colorado.edu/en/simulations/states-of-matter>

Curriculum links: AC9S7U05 · AC9S8U06

Learning intentions

- I can describe how particle motion and spacing differ between solids, liquids and gases.
- I can relate heating and cooling to changes of state at the particle level.
- I can classify a substance as an element, compound or mixture based on its particle representation.

Getting started

Open PhET States of Matter and choose a substance, starting with a simple one like neon or oxygen. Use the heater/cooler control at the bottom of the screen and watch the particles in the main view as you change the temperature.

Tasks

1. Start with your substance as a solid. Describe how the particles are arranged and how they are moving.

2. Slowly heat the substance until it becomes a liquid. What changes about the particle spacing and motion?

3. Keep heating until the substance becomes a gas. Describe the particle behaviour now, compared with the solid and liquid stages.

4. Record the particle spacing and motion you observed for each of the three states.

State	Particle spacing	Particle motion

5. Explain, in terms of particle motion and attraction, why a substance expands as it changes from solid to liquid to gas.

6. Switch to the Phase Diagram tab, if available, and locate the point representing your substance's current temperature and pressure. What state does the diagram predict?

7. Using the particle view, explain the difference between how you would represent an element compared with a compound in this simulation.

Extension (optional)

Predict what would happen to the phase diagram of a substance if you increased the pressure while keeping temperature constant. Test your prediction in the simulation and explain the result.
